

Of Scales and Displacement: the Need for a New Quantum Law

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Abstract

In this note, the author calls attention to the need for a new law of combining velocities of two different types: scale change velocity and the usual Galilean displacement associated velocity. This law is more relevant in the quantum domain than in the classical domain.

Consider an observer in a sea of information, as posited in the figure in the paper titled “Information is Fundamental, not Space or Time”. This observer could be ourselves. Suppose now that this observer is in motion. We will assume that the motion is ascertained with reference to time and time is a fictional fulcrum for the mass of ideas within which we operate.

Motion with respect to a background means he or she has a velocity, v that can be computed. As he or she wades through the sea of information, they would encounter information at different densities ρ , depending on two factors:

1. their own velocity,
2. the actual isotropy or non-isotropic spread of information in the environment.

With respect to point (2.) above, information is spread in a non-homogeneous way in the environment. The homogeneity or lack thereof is ascertained with respect to the spatiotemporal scale on which the observer is operating. At the nano scale there is a high density of information and the same setting when viewed at the macro level appears to be characterless and devoid of information. Thus the motion could be a scale change. The observer could be ascending from the nano to the macro or descending from the micro to the nano, for example. So during the ascent (or descent), density goes down (or up, respectively) in time¹.

Taking into account point (1.) above, we may want to write an expression such as:

¹However, we may also want to consider a fixed scale. Then, density may vary with space at that very temporal scale. This is the usual Galilean case. In this context we will denote the corresponding velocity as $v_{Galileo}$, to contrast it with v_{scale} .

$$\rho(I(t_2)) - \rho(I(t_1)) = f(v). \quad (1)$$

That is to say, the difference in the “density” of the information encountered at time t_1 and time t_2 , is proportional to a function f of the speed or velocity v of the observer. According to this formula, if the velocity is 0, then the observer will see no difference or change in the density of information at t_2 versus at t_1 . It will appear to be a homogeneous, isotropic, steady-state distribution of information. The other way around, from the degree of non-homogeneity he can infer his own velocity, either v_{scale} or $v_{Galileo}$. Thus, using this equation, we have recovered the old, known, relation between spacetime and velocity, under certain assumptions. We assume that ρ is a linear function of its arguments. Then

$$\rho(I(t_2)) - \rho(I(t_1)) = \rho(I(t_2) - I(t_1)) = \rho(\Delta I) = \rho(g(\Delta t)) = h(\Delta t) = f(v). \quad (2)$$

That is to say, the velocity is a function of the change in time and change in space – the classical Galilean relation (see for example [1]). Thus, what we may have here, is an actual, bona-fide generalization of classical mechanics. Instead of generalizing along the relativistic direction, we have generalized along the information theoretic direction. Further, we may be able to always assume the case is point (1.) above and not (2.), for in an absolute or ideal sense, there may be no variation in the spread of information in space and time – the density may be everywhere the same. This can probably be taken as a postulate - that v_{scale} is more fundamental than $v_{Galileo}$ since the latter contributes minimally to variation in information distribution.

Consider however what happens when the observer has a velocity which is a combination of the two above. Will the usual laws of combining velocity vectors of different directions, dimensions, or subspaces apply here? It is unlikely. With regards to Equation (2) above, the relevant velocity might be weighted more dramatically towards v_{scale} than influenced by $v_{Galileo}$, for a very minute change in scale position would make an immediate and dramatic difference to the perceived spread or density of information content.

As a matter of fact, an observer stationary with respect to the Earth’s surface, almost always has a $v_{Galileo}$. He may or may not have a v_{scale} . It is clear however, if that observer is experiencing dramatic changes in information density, then scale changes must be taking place, ever so minute or large as well.

The question becomes: are there any quantum domain living organisms? For if there were, they could experience scale change along with significant Galilean change. As one comes up to the classical domain, Galilean contribution to the net information density change becomes negligible. So if one descends to the quantum domain (as a specific organism) and this Galilean contribution increases in its magnitude, then we would need a new law for combining the two velocities.

Note that our discussion assumes that Sankara’s equation of scale invariance (*atman=brahman*) applies to all beings, regardless of the scale at which they

primarily reside. Scales can also be thought of as the various Sanskrit “*loka*”s or planes of existence.

References

- [1] Eizo Nakaza. Resolving our erroneous interpretation of the galilean transformation. *Physics Essays*, 28(4):503–506, 2015.